Notes: Kugler & Verhoogen (RES, 2012)

Prices, Plant Size, and Product Quality

Contents

1 Big picture

- **Question.** Why do larger manufacturing plants charge higher output prices and pay higher input prices within narrowly defined product categories? Are these price-size patterns evidence of product quality, input quality, or market power?
- **Setting.** The paper uses the Colombian manufacturing census, an unusually rich plant-level data set with physical quantities and values for both outputs and material inputs. This lets the authors construct unit values for narrowly defined products at the plant-product-year level.
- **Main empirical facts.** Within product-year cells, larger plants charge higher output unit values and pay higher input unit values. Exporters show similar patterns: they charge higher output prices and pay higher input prices than non-exporters within the same product-year categories.
- **Main theoretical idea.** The authors extend the Melitz heterogeneous-firm model by allowing firms to choose both input quality and output quality. More capable plants either have a comparative advantage in using high-quality inputs or can spread fixed quality-upgrading costs over larger sales volumes.
- **Core prediction.** If the scope for quality differentiation is large, more capable plants choose higher-quality inputs, pay more for inputs, produce higher-quality outputs, and charge higher output prices. The price-size elasticities should be stronger in sectors where quality differentiation is more important.
- **Cross-sector test.** The paper uses U.S. industry R&D plus advertising intensity, following Sutton, as an off-the-shelf proxy for the scope for quality differentiation. Output and input price-size elasticities are larger in sectors with higher R&D/advertising intensity.
- **Alternative explanations.** Market power could in principle generate price-size correlations. The paper tests supplier concentration, purchaser concentration, purchaser share, horizontal differentiation, and Rauch measures. These do not fully explain the facts.
- **Economic takeaway.** Quality upgrading appears to involve both the products a firm sells and the inputs it buys. Industrial upgrading may require upgrading whole supplier-buyer complexes, not only upgrading isolated lead firms.

1.1 Incremental contribution of the paper

- **Product-level evidence inside plants.** Earlier heterogeneous-firm research mainly used plant revenues, wages, exports, and measured productivity. This paper brings the product dimension into the plant-level literature by using physical quantities and values for individual outputs and inputs.
- **Input prices, not just output prices.** A positive output price-size relation could reflect markups, demand shocks, or quality. The paper's incremental contribution is to show that larger firms also pay higher prices for material inputs within narrow categories, which is much harder to rationalize with output-market demand shocks alone.
- **Generalization of the size-wage premium.** The paper extends the employer size-wage effect from labor to material inputs: larger plants appear to buy more expensive, plausibly higher-quality inputs, not merely to pay higher wages.
- **Quality complementarity in a Melitz environment.** The model connects plant capability, input quality, output quality, and export selection in a tractable heterogeneous-firm framework. This gives a microfoundation for why more productive or larger plants may have higher observed prices rather than lower prices.
- **Cross-sector validation.** The paper does not stop at average correlations. It tests whether price-size elasticities are stronger in industries where quality differentiation is ex ante more important, using Sutton-style R&D/advertising intensity.
- **Discriminating quality from market power.** By constructing supplier Herfindahl, purchaser Herfindahl, purchaser share, Gollop-Monahan differentiation, and Rauch classifications, the paper directly evaluates competing market-power interpretations.

2 Data construction and measurement

2.1 Colombian manufacturing census

- The main data source is the Colombian annual manufacturing census, covering manufacturing plants with at least 10 workers.
- The main period is 1982–2005. The export-status analysis uses 1982–1994 because export status is consistently observed in those years.
- The data are representative of formal manufacturing and include plant identifiers, employment, production values, sales, inventories, output product codes, input product codes, physical quantities, and values.
- The key advantage is that the paper observes both the numerator and denominator of unit values: value and physical quantity. This allows the authors to compute output and input prices at the plant-product-year level.

Interpretation: The empirical design uses variation across plants producing or buying the same product in the same year. This avoids comparing pesos per kilogram of one good with pesos per unit of another good.

2.2 Output prices and input prices

- The output price is the log real unit value of a product produced by a plant: value of production divided by physical quantity.
- The input price is the log real unit value of a material input purchased by a plant: expenditure divided by physical quantity.
- Product-year fixed effects absorb all economy-wide and product-specific price changes for a given product in a given year.
- Region-year fixed effects absorb regional transport-cost differences and local shocks.
- Industry effects absorb broad sectoral differences in plant technologies and product mixes.

Interpretation: Unit values are imperfect prices because they may include within-product quality variation. That imperfection is exactly what the paper exploits: if unit values differ across plants within the same narrow product-year cell, the residual differences are informative about quality, markups, or measurement error.

2.3 Plant size and export variables

- Plant size is measured mainly by log employment and log total output.
- Total output is the value of production: sales plus net transfers plus net change in inventories.
- Log employment is useful because it is less mechanically related to unit values than revenue-based output.
- In 2SLS specifications, log employment is used as an instrument for log total output to mitigate measurement error in output.
- Exporter is an indicator for positive exports. Export share is exports divided by total sales.

Interpretation: The distinction between employment and revenue is important. Regressing prices on revenue can mechanically induce a positive correlation because revenue contains price. Employment provides a cleaner measure of scale.

2.4 Industry-level scope for quality differentiation

- The Colombian data do not contain plant-level R&D and advertising expenditures.
- The paper uses U.S. FTC Line of Business Survey data on industry R&D plus advertising intensity as a proxy for the scope for quality differentiation.
- The logic follows Sutton: industries with large R&D and advertising outlays are industries where firms can use fixed outlays to improve actual or perceived quality.
- The paper maps U.S. industry measures into Colombian manufacturing sectors through concordances.

Interpretation: Using U.S. R&D/advertising intensity as an external industry characteristic reduces the risk that the quality-differentiation proxy is itself mechanically generated by Colombian plant prices.

2.5 Horizontal differentiation and market-power variables

- The Gollop-Monahan index measures horizontal differentiation using dissimilarity in input mixes across plants within industries.
- The Rauch measure classifies goods as homogeneous/reference-priced or differentiated.
- Supplier Herfindahl is the sum of squared market shares of producers of a given input.
- Purchaser Herfindahl is the sum of squared expenditure shares of plants purchasing a given input.
- Purchaser share is a plant's share of total domestic expenditure on an input in a given year.

Interpretation: These variables are designed to separate the quality story from pricing-to-firm, bargaining, and monopsony stories. If market power were the sole explanation, input price-size elasticities should be systematically stronger where suppliers or buyers have more market power.

3 Model and empirical specifications

3.1 Baseline price-size regression

The baseline empirical specification is:

$$\ln p_{ijt} = X_{jt}\gamma + \psi_{it} + \delta_{rt} + \xi_k + \varepsilon_{ijt}. \tag{1}$$

Here i indexes products, j plants, k industries, and t years. The dependent variable is the log real unit value of an output or an input. X_{jt} is plant size, export status, or export share. ψ_{it} is a product-year fixed effect, δ_{rt} is a region-year fixed effect, and ξ_k is an industry effect.

Interpretation: The coefficient γ compares prices across plants of different size that produce or use the same product in the same year. The coefficient is a correlation, not a causal effect of plant size on price. The authors interpret it as evidence that unobserved plant capability is correlated with input and output quality choices.

3.2 Measurement-error and 2SLS specification

A key issue is that gross output is revenue-based and therefore contains prices. To reduce mechanical bias, the paper uses log employment as a reduced-form size measure and as an instrument for log total output:

$$\ln(\text{output}_{jt}) = \pi_0 + \pi_1 \ln(\text{employment}_{jt}) + \text{fixed effects} + u_{jt}. \tag{2}$$

The second stage replaces plant size with the fitted component of log total output.

Interpretation: If unit values are measured with error, revenue and unit values can share measurement error. Employment helps break this mechanical link. The similarity of OLS, reduced-form, and 2SLS results strengthens the stylized facts.

3.3 Export-status specification

For the 1982–1994 panel, the authors estimate the same price equation with exporter status or export share:

$$\ln p_{ijt} = \beta_1 \text{Exporter}_{jt} + \beta_2 \text{ExportShare}_{jt} + \psi_{it} + \delta_{rt} + \xi_k + \varepsilon_{ijt}. \tag{3}$$

Interpretation: Exporters are larger and more capable in standard heterogeneous-firm models. If capability is also related to quality choices, exporters should charge and pay higher prices even within product-year cells.

3.4 Quality-augmented Melitz model: demand and production

Consumers have CES preferences over quality-adjusted varieties:

$$U = \left[\int_{\omega \in \Omega} (q(\omega) x(\omega))^{\frac{\sigma-1}{\sigma}} \, d\omega \right]^{\frac{\sigma}{\sigma-1}}. \tag{4}$$

Demand for a variety is increasing in its quality $q(\omega)$ and decreasing in its output price $p_O(\omega)$.

Intermediate inputs of quality c are produced with:

$$F_I(\ell, c) = \frac{\ell}{c}, \qquad p_I(c) = c. \tag{5}$$

Final-good physical output is:

$$F(n) = n\lambda^a, \tag{6}$$

where λ is plant capability and $a > 0$ governs the cost advantage of capability.

Interpretation: The key modeling move is to make input quality costly and observable. Higher-quality inputs cost more. Therefore, if larger plants buy higher-quality inputs, input prices rise with size.

3.5 Variant 1: input quality and plant capability are complements

The first variant assumes:

$$q = \left[\frac{1}{2}\lambda^{b\theta} + \frac{1}{2}c^{2\theta} \right]^{1/\theta}, \qquad \theta < 0. \tag{7}$$

This implies log-supermodularity between plant capability and input quality. The equilibrium choices are:

$$c^*(\lambda) = p_I^*(\lambda) = \lambda^{b/2}, \tag{8}$$

$$q^*(\lambda) = \lambda^b, \tag{9}$$

$$p_O^*(\lambda) = \frac{\sigma}{\sigma - 1}\lambda^{b/2-a}. \tag{10}$$

Interpretation: More capable plants have a comparative advantage in transforming better inputs into better outputs. Input prices rise with capability whenever $b > 0$. Output prices rise with capability when the quality effect is large enough to offset the cost-reduction effect.

3.6 Variant 2: fixed costs of quality upgrading

The second variant assumes quality requires both fixed quality investment and high-quality inputs:

$$q = \min\{f_q^{\alpha}, c^2\}. \tag{11}$$

The parameter α measures the scope for quality differentiation. The model implies that more capable plants can spread fixed quality costs over larger output volumes and therefore choose higher f_q, higher input quality, and higher output quality.

Interpretation: This variant is closest to Sutton's escalation logic. Industries with high R&D and advertising intensity are those in which fixed outlays can raise quality more effectively; hence price-size elasticities should be stronger in such industries.

3.7 Cross-sector interaction regressions

The key empirical test of the model is:

$$\ln p_{ijt} = \gamma_1 \ln L_{jt} + \gamma_2 (\ln L_{jt} \times QD_k) + \psi_{it} + \delta_{rt} + \xi_k + \varepsilon_{ijt}, \tag{12}$$

where QD_k is R&D plus advertising intensity or another differentiation measure.

Interpretation: A positive γ_2 means the price-size relationship is steeper in sectors where quality differentiation is easier or more important. This is the main cross-sector test of the quality interpretation.

4 Identification and empirical strategy

4.1 What is identified

The empirical estimates identify conditional correlations between prices and plant size within narrow product-year cells. The paper does not claim that making a plant larger causes it to choose higher prices. Instead, the theory interprets plant size, input quality, output quality, and prices as joint outcomes of latent plant capability.

4.2 Why product-year fixed effects matter

Product-year fixed effects force comparisons to be within the same product and year. This removes aggregate inflation, common product price shocks, and product-unit heterogeneity. It also avoids the difficulty of comparing physical unit values across different goods.

4.3 Why input prices are central to identification

A positive output price-size correlation could be caused by quality, market power, or plant-specific demand. Input prices provide an additional diagnostic. If larger plants pay higher prices for the same input category, the most natural interpretation is that they buy higher-quality inputs, unless input-market market power or bargaining can explain the pattern.

4.4 Cross-sector quality test

The quality model predicts not only positive average correlations but also stronger correlations in sectors with greater scope for quality differentiation. The use of external U.S. R&D/advertising intensity makes the test sharper because the sector-level proxy is not built from Colombian price outcomes.

4.5 Market-power falsification strategy

The paper considers two main alternatives. First, suppliers with market power might price discriminate against larger or less elastic plants. Second, large downstream plants might move up upward-sloping input supply curves. Supplier and purchaser Herfindahl indices, purchaser share, Gollop-Monahan measures, and Rauch classifications are used to test these channels.

4.6 Threats and limitations

- Unit values are imperfect prices because product codes may still contain quality heterogeneity.
- Product quality is not directly observed, so the quality interpretation is indirect.
- Plant-level firms are not observed; the analysis treats plants as single-establishment firms.
- R&D/advertising intensity comes from U.S. industry data and must be mapped into Colombian sectors.
- The results are correlations and do not isolate causal changes in plant capability, product quality, or input quality.

Interpretation: The identification strategy is a triangulation strategy. No single regression proves quality. The case rests on the combination of input and output prices, within-product comparisons, cross-sector quality proxies, and negative evidence against market-power alternatives.

5 Tables and figures

5.1 Figure 1: Illustrative example: Hollow Brick + Figure 2: Illustrative example: Bar Soap

Read:

- Figure 1 shows hollow brick, a product with plausibly limited scope for quality differentiation. Both the output price-employment slope and the input price-employment slope are close to zero.
- Figure 2 shows bar soap, a product with clear scope for quality differentiation through brands, composition, odor, purity, and consumer perceptions. The output price and both input price slopes are positive.
- The contrast previews the paper's central hypothesis: price-size correlations are stronger when quality differentiation is technologically or commercially important.

Interpretation: The hollow-brick example resembles the standard Melitz intuition: larger plants may be more efficient but do not necessarily charge higher observed physical-unit prices. The bar-soap example fits the quality-upgrading interpretation: larger plants produce more expensive outputs and buy more expensive inputs.

Economic message: The difference between the two examples is not just noise. It is the visual foundation for the cross-sector test: sectors with more quality scope should have steeper output and input price-size gradients.

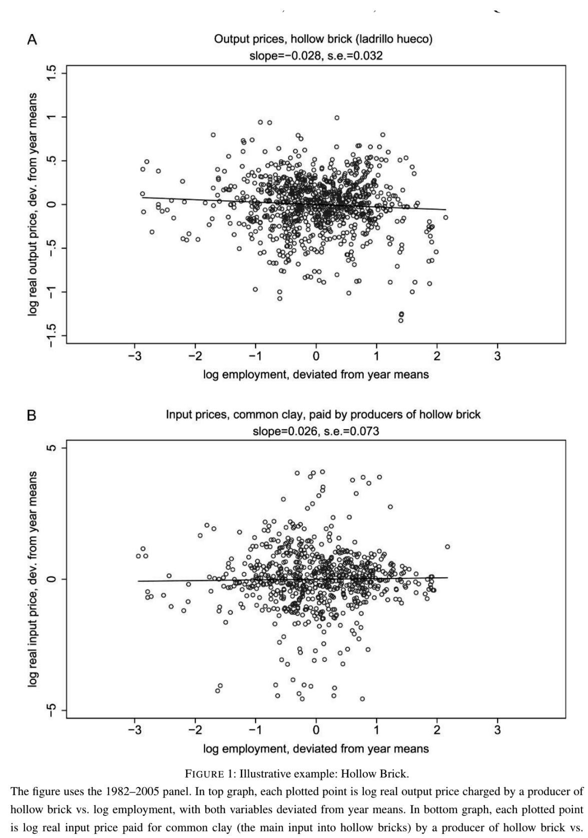

Figure 1: Illustrative example: Hollow Brick

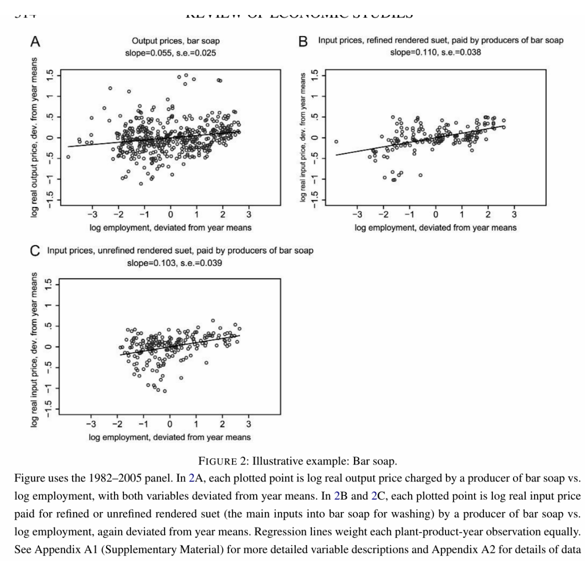

Figure 2: Illustrative example: Bar Soap

5.2 Table 1: Product prices vs. plant size + Table 2: Product prices vs. export measures, plant size

Read:

- Table 1 shows positive average relationships between plant size and output/input prices. A 10% increase in employment is associated with about 0.26% higher output prices and about 0.12% higher input prices.
- The 2SLS results using log employment as an instrument for log output preserve the positive output and input price relationships.
- Table 2 shows that exporters charge higher output prices and pay higher input prices. Exporter status and export share are positively related to output prices; exporter status is also positively related to input prices.

Interpretation: Table 1 establishes the basic stylized facts using the full panel. Table 2 shows that the same quality-related patterns appear in the export margin. If exporters are more capable plants, their higher output and input prices are consistent with capability-linked quality upgrading.

Economic message: These tables challenge the simple interpretation that larger firms are just lower-cost producers charging lower prices. Larger plants appear to operate in higher-quality input-output space.

TABLE 1
Product prices vs. plant size

	OLS (1)	Reduced form (2)	2SLS (3)
A. Dependent variable: log real output unit value			
Log total output	0·021***		0·025***
	(0·005)		(0·006)
Log employment		0·026***	
		(0·007)	
Product-year effects	Y	Y	Y
Industry effects	Y	Y	Y
Region-year effects	Y	Y	Y
R^2	0·90	0·90	
N (obs.)	413,789	413,789	413,789
N (plants)	13,582	13,582	13,582
B. Dependent variable: log real input unit value			
Log total output	0·015***		0·011***
	(0·002)		(0·003)
Log employment		0·012***	
		(0·003)	
Product-year effects	Y	Y	Y
Industry effects	Y	Y	Y
Region-year effects	Y	Y	Y
R^2	0·78	0·78	
N (obs.)	1,338,921	1,338,921	1,338,921
N (plants)	13,582	13582	13,582

Notes: The table uses the 1982–2005 panel. Total output is total value of production, defined as sales plus net transfer plus net change in inventories. In Column 3, log employment is instrument for log total output; the coefficient on lo employment, its robust standard error and the R^2 in the first stage are 1·058, 0·011, and 0·733, in Panel A and 1·082 0·10 and 0·782 in Panel B, respectively. Product-year and industry effects are not perfectly collinear because industry i defined as the industry category with the greatest share of plant sales, and two plants producing the same product may b

Table 1: Product prices vs. plant size

TABLE 2
Product prices vs. export measures, plant size

	(1)	(2)	(3)	(4)	(5)
A. Dependent variable: log real output price					
Log employment	0·025***			0·009	0·020**
	(0·008)			(0·008)	(0·008)
Exporter		0·114***		0·104***	
		(0·022)		(0·023)	
Export share			0·288**		0·251*
			(0·137)		(0·142)
Product-year effects	Y	Y	Y	Y	Y
Industry effects	Y	Y	Y	Y	Y
Region-year effects	Y	Y	Y	Y	Y
R^2	0·90	0·90	0·90	0·90	0·90
N (obs.)	216,155	216,155	216,155	216,155	216,155
N (plants)	10,106	10,106	10,106	10,106	10,106
B. Dependent variable: log real input price					
Log employment	0·013***			0·008**	0·013***
	(0·004)			(0·004)	(0·004)
Exporter		0·037***		0·028***	
		(0·009)		(0·009)	
Export share			0·021		−0·002
			(0·027)		(0·027)
Product-year effects	Y	Y	Y	Y	Y
Industry effects	Y	Y	Y	Y	Y
Region-year effects	Y	Y	Y	Y	Y
R^2	0·80	0·80	0·80	0·80	0·80
N (obs.)	684,746	684,746	684,746	684,746	684,746
N (plants)	10,106	10,106	10,106	10,106	10,106

Notes: The table uses the 1982–1994 panel since export status is reported on a consistent basis only for those years. Exporter equals 1 if plant has exports >0, and 0 otherwise. Export share is fraction of total sales derived from exports.

Table 2: Product prices vs. export measures, plant size

5.3 Figure 3: Price-plant size elasticities vs. R&D/advertising intensity + Table 3: Interactions with differentiation measures for output sectors

Read:

- Figure 3 plots sector-specific price-employment elasticities against R&D/advertising intensity. The fitted lines slope upward for both output and input prices.
- Table 3 formalizes the figure by interacting log employment with R&D/advertising intensity and other differentiation measures.
- For output prices, the interaction with R&D/advertising intensity is positive and statistically significant. For input prices, the interaction is also positive in the R&D/advertising columns.

- The Gollop-Monahan and Rauch controls address the concern that the result is simply horizontal differentiation rather than vertical quality differentiation.

Interpretation: This is the paper's key model test. The model predicts that price-size gradients should be steeper in sectors where quality upgrading is more valuable. The evidence is consistent with that prediction, especially for the R&D/advertising measure.

Economic message: The result gives cross-sector content to the quality story. It is not just that some plants have high unit values; the pattern is strongest precisely where quality ladders are expected to matter.

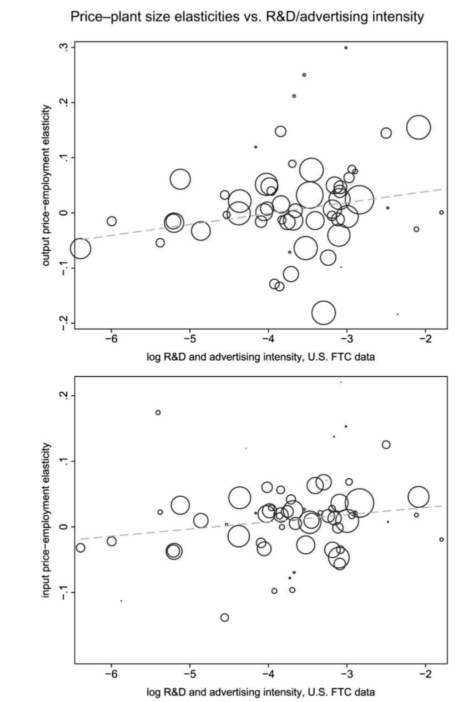

FIGURE 3: Price-plant size elasticities vs. R&D/advertising intensity.
ecific output (input) price–employment elasticities calculated from a regression of log real output
log employment interacted with four-digit sector dummies, with product-year, region-year and fi
iies as additional covariates. Fitted regression line is weighted by total output in each industry.
by sizes of circles. To increase legibility, the following outliers (with elasticity estimates in paren
tted from the graph: 3833 (−0·41), 3530 (−0·38), 3821 (0·72), 3841 (0·83), 3842 (2·46) for outpu
3852 (−0·33), 3530 (0·17), 3853 (0·22) for inputs. These sectors (which have few observations an
regression results) are included in the regressions reported in Table 3. To map plotted circles to in

Figure 3: Price-plant size elasticities vs. R&D/advertising intensity

TABLE 3
Interactions with differentiation measures for output sectors

	Dependent variable: log real output price				Dependent variable: log real input price			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log employment	0·029***	0·029***	0·028***	0·028***	0·013***	0·013***	0·013***	0·013***
	(0·007)	(0·007)	(0·007)	(0·007)	(0·003)	(0·003)	(0·003)	(0·003)
Log emp.*log(R&D + adv. intensity)		0·022**	0·022**	0·019**		0·010**	0·009*	0·010**
		(0·009)	(0·009)	(0·009)		(0·005)	(0·005)	(0·005)
Log emp.*Gollop-Monahan index			0·126***				0·051**	
			(0·035)				(0·024)	
Log emp.*Rauch index				0·045***				−0·004
				(0·015)				(0·009)
Product-year effects	Y	Y	Y	Y	Y	Y	Y	Y
Industry effects	Y	Y	Y	Y	Y	Y	Y	Y
Region-year effects	Y	Y	Y	Y	Y	Y	Y	Y
R^2	0·90	0·90	0·90	0·90	0·79	0·79	0·79	0·79
N (obs.)	341,812	341,812	341,812	341,812	1,133,366	1,133,366	1,133,366	1,133,366
N (plants)	12,356	12,356	12,356	12,356	11,406	11,406	11,406	11,406

Notes: Data on advertising and R&D expenditures as a share of total industry sales are from the U.S. FTC 1975 Line of Business Survey, converted from FTC four-digit industry classification to ISIC four-digit rev. 2 classification using verbal industry descriptions. At SITC four-digit level, Rauch (1999) measure is set to 0 if good is "homogeneous" or "reference-priced" according to the Rauch "liberal" definition, to 1 if reported not to be in either category, and then concorded to ISIC rev. 2 four-digit categories. Differentiation measures have been deviated from global means. The table uses the 1982–2005 panel. Sample includes sectors with complete data on advertising and R&D intensity, Gollop–Monahan index and Rauch measures, which excludes a small number of sectors. Columns 1 and 6 correspond to Column 2 of Table 1 but use reduced sample. Errors clustered at plant level. *N* (plants) reports number of clusters (*i.e.* distinct plants that appear in any year). Four-digit industry for outputs (Columns 1–4) defined as first four digits from eight-digit product code. Four-digit industry for inputs (Columns 5–8) defined as industry of corresponding plant. Number of plants differ between Columns 1–4 and 5–8 because of difference in industry definitions. Robust standard errors are given in parentheses. *10% level, **5% level, ***1% level. See Appendix A1 for more detailed variable descriptions and Appendix A2 for details of data processing.

Table 3: Interactions with differentiation measures for output sectors

5.4 Table 4: Interactions with differentiation measures for input sectors + Table 5: Interactions with market power measures for input sectors

Read:

- Table 4 assigns differentiation measures to input sectors. It shows that the input price-size elasticity is stronger in input sectors with higher R&D/advertising intensity and is also related to the Rauch differentiation measure.
- Table 5 tests market-power mechanisms. The supplier Herfindahl interaction is negative, not positive, which runs against a simple supplier-market-power explanation.
- Purchaser share is positive, suggesting some role for monopsony or large-buyer position, but the basic input price-size relation remains even when market-power controls are included.

Interpretation: Table 4 reinforces the quality interpretation on the input side: larger plants pay more for inputs especially where input quality differentiation is plausible. Table 5 shows that market power is not a complete substitute explanation. Some bargaining or purchaser-power mechanisms may matter, but they do not eliminate the quality pattern.

Economic message: The two tables together are important because input prices are the hardest part of the story for pure demand or markup explanations. The evidence suggests that larger firms buy better inputs, not merely that they charge higher output markups.

TABLE 4
Interactions with differentiation measures for input sectors

	Dependent variable: log real input price			
	(1)	(2)	(3)	(4)
Log employment	0·013***	0·013***	0·013***	0·013***
	(0·003)	(0·003)	(0·003)	(0·003)
Log emp.*log(R&D + advertising intensity, input sector)		0·008***	0·008***	0·005*
		(0·003)	(0·003)	(0·003)
Log emp.*GM index (input sector)			0·008	
			(0·011)	
Log emp.*Rauch index (input sector)				0·026***
				(0·008)
Product-year effects	Y	Y	Y	Y
Industry effects	Y	Y	Y	Y
Region-year effects	Y	Y	Y	Y
R^2	0·76	0·76	0·76	0·76
N (obs.)	1,037,055	1,037,055	1,037,055	1,037,055
N (plants)	13,128	13,128	13,128	13,128

Notes: Differentiation measures are assigned to input sectors, defined as first four digits from eight-digit product code fo corresponding input. Sample includes sectors with complete data on advertising and R&D intensity and Rauch measures Data on advertising and R&D expenditures as a share of total industry sales are from the U.S. FTC 1975 Line of Busines Survey, converted from FTC four-digit industry classification to ISIC four-digit rev. 2 classification using verbal industr descriptions. At SITC four-digit level, Rauch (1999) measure set to 0 if good is "homogeneous" or "reference-priced according to the Rauch "liberal" definition, to 1 if reported not to be in either category, and then concorded to ISIC rev 2 four-digit categories. Differentiation measures have been deviated from global means. The table uses the 1982–200 panel. Sample includes sectors with complete data on advertising and R&D intensity, Gollop–Monahan (GM) index an Rauch measures for input sectors, which excludes a small number of sectors. Column 1 corresponds to Column 2, Pane B of Table 1 but uses reduced sample. Errors clustered at plant level. *N* (plants) reports number of cluster (*i.e.* distinc plants that appear in any year). Robust standard errors are given in parentheses. *10% level, **5% level, ***1% level See Appendix A1 for more detailed variable descriptions and Appendix A2 for details of data processing.

Table 4: Interactions with differentiation measures for input sectors

TABLE 5
Interactions with market power measures for input sectors

	Dependent variable: log real input unit value					
	(1)	(2)	(3)	(4)	(5)	(6)
Log employment	0·012***	0·019***	0·010*	0·010***	0·009***	0·018***
	(0·003)	(0·004)	(0·006)	(0·003)	(0·003)	(0·007)
Log emp.*Herfindahl index (suppliers)		−0·015**				−0·019***
		(0·006)				(0·006)
Log emp.*GM index (input sector)			0·004			−0·000
			(0·011)			(0·011)
Log emp.*Herfindahl index (purchasers)				0·016		−0·003
				(0·011)		(0·011)
Purchaser share					0·240***	0·249***
					(0·037)	(0·038)
Product-year effects	Y	Y	Y	Y	Y	Y
Industry effects	Y	Y	Y	Y	Y	Y
Region-year effects	Y	Y	Y	Y	Y	Y
R^2	0·76	0·76	0·76	0·76	0·76	0·76
N (obs.)	1,050,029	1,050,029	1,050,029	1,050,029	1,050,029	1,050,029
N (plants)	13,285	13,285	13,285	13,285	13,285	13,285

Notes: Herfindahl index of suppliers is sum of squared market shares of producers of input, at eight-digit industry level Herfindahl index of purchasers is sum of squared expenditure shares of purchasers of input, at eight-digit industry level The Gollop–Monahan (GM) index (input sector) is assigned based on industry of inputs. The purchaser share defined a expenditures on product by plant as a share of total expenditures on product by all plants in a given year. Differentiatio and concentration measures have not been deviated from global means (unlike in Tables 3 and 4). The table uses th 1982–2005 panel. Sample includes plant-product-year observations for which both Herfindahl indices, GM index an purchaser share could be constructed; plants that use only non-manufacturing inputs are excluded. Errors clustered a plant level. *N* (plants) reports number of clusters (*i.e.* distinct plants that appear in any year). Robust standard errors ar given in parentheses. *10% level, **5% level, ***1% level. See Appendix A1 for more detailed variable description

Table 5: Interactions with market power measures for input sectors

6 Interpretive synthesis

- **Average facts.** Larger plants and exporters have higher output and input unit values within narrow product-year categories.
- **Theory.** A quality-augmented heterogeneous-firm model can produce these facts if more capable plants choose higher-quality inputs and outputs.
- **Cross-sector validation.** The facts are stronger in industries where external measures indicate more scope for quality differentiation.
- **Alternative explanations.** Market power can explain some patterns but does not provide a full account, especially on the input side.
- **Development implication.** Quality upgrading is relational. The ability of downstream firms to upgrade may depend on access to high-quality inputs, and the incentives of suppliers to upgrade may depend on demand from capable downstream firms.

7 Short summary

- The paper uses Colombian manufacturing census data with plant-product-level input and output unit values.
- Within narrow product-year cells, larger plants charge higher output prices and pay higher input prices.
- Exporters show similar patterns: they charge and pay higher prices than non-exporters.

- A standard Melitz model cannot easily explain positive input price-size correlations.
- The authors extend Melitz with endogenous input and output quality choices.
- In both model variants, more capable plants choose higher-quality inputs and outputs when the scope for quality differentiation is positive.
- Output prices rise with size when the quality effect dominates the cost-reduction effect.
- The model predicts stronger price-size elasticities in sectors with greater scope for quality differentiation.
- R&D plus advertising intensity is used as a proxy for this scope, following Sutton.
- The data support the prediction for both outputs and inputs.
- Market power measures do not fully explain the observed price-size patterns.
- The main conclusion is that input quality and output quality are central to understanding plant heterogeneity.

8 Conclusion

8.1 Core conclusion

Kugler and Verhoogen show that larger plants do not merely produce more; they also operate at different points in the quality distribution. They charge more for outputs and pay more for inputs within narrow product categories, and these relationships are stronger in sectors where quality differentiation is more important.

8.2 Economic intuition

A more capable plant has two opposing forces. Higher capability lowers production costs, which by itself would reduce prices. But higher capability also raises the return to upgrading output quality and input quality. If quality differentiation is sufficiently valuable, the quality channel dominates and observed physical-unit prices rise with plant size.

8.3 What the paper adds to heterogeneous-firm trade

The paper shifts attention from productivity alone to the joint choice of productivity, input quality, output quality, and market participation. It also shows that exporters' higher prices need not be puzzling: exporting can be associated with higher capability and higher product quality.

8.4 Limits of the evidence

The paper does not observe quality directly, and the empirical regressions are correlations. The evidence is strongest as a triangulation exercise: input and output prices, export status, sectoral quality proxies, and market-power tests all point toward quality differences as a central explanation.

8.5 Takeaway

The main lesson is that industrial upgrading is not only about producing more cheaply. It is also about moving into higher-quality outputs and using higher-quality inputs. The supply chain matters: high-quality downstream production may require high-quality upstream inputs, and demand from capable firms may induce upstream upgrading.